Artificial Intelligence versus Food Wastage in Bakeries

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Abstract. Time series forecasting for sales is well established in the business world. However, the nature of the data and often the domain area of the application can yield unexpected inconsistencies and constraints. Machine learning tools can be utilized to enhance the accuracy of predictions and support effective forecasting. This work presents the case of a small chain of bakeries where machine learning supported forecasting aims to tackle food wastage, enhance profit margins, and contribute to a more sustainable community.

Keywords: Machine Learning, Sales forecasting, Time Series Forecasting, Bakeries, Food Wastage, Sustainability.

1 Introduction

In January 2025 the United Kingdom government announced its AI Opportunities Plan, aiming at ramping up AI adoption across the UK to boost economic growth, provide jobs for the future and improve people's everyday lives [1]. While the Department of Science, Innovation and Technology of the UK Government is developing its long-term plan for the UK's AI infrastructure needs, backed by a 10-year investment commitment, funding initiatives target more specific needs within the business sectors that could have a more immediate impact on society. Small businesses across the country will be helped to boost their productivity and efficiency through AI tools set to be trialed. In January 2025 the government announced the launch of 120 funded projects to support small businesses in engaging AI to enhance their efficiency. Among those projects there are ones supporting bakers to use AI to cut food waste – and thereby protect profit margins. The aim is to accurately predict sales and forecast how much each product needs to be made daily [2]. Minimizing food wastage could induce lower raw material demands that could eventually lead to price drops in the medium to long-term, allowing households and individuals a stronger purchase power for foods that provide better nutrition. Beyond the business impact such projects have a strong societal and sustainability focus. The United Kingdom, despite being one of the world's leading economies, is seeing an alarmingly increasing number of people relying on food banks to feed themselves and their families [3]. Food wastage can be seen as accentuating this problem as it contravenes societal ethics. At the same time, such projects can be seen as conversant with

the United Nations (UN) sustainability Goals that were set in 2015. Among the 17 Goals there is a strong focus on protecting vulnerable individuals, with Goal 2 targeting hunger and more specifically aims at ending hunger by achieving food security and nutrition, and sustainable agriculture. The UN recognizes that high food prices are a key factor in driving communities and whole countries to hunger [4]. Combating and controlling food wastage can help control food price rises and supporting communities addressing huger issues.

The work discussed here is based on a project working with a small bakery chain in Bristol, United Kingdom, to provide stable prediction of demand for products that have a short shelf life. Accurate prediction can reduce food waste and improve profit margins within the company. The first phase of the work that is discussed here was carried out between August 2024 and March 2025. This phase of the work was funded by the Made Smarter West of England initiative [5]. The work is ongoing with the next phase planned for the summer of 2025.

2 The nature of data at the core of the case study

Artificial Intelligence tools are usually based on having access to large volumes of data that allow for fine-tuned forecasting via stable time series. Bakeries, like any retail business, produce sales data that are in principle suitable for applying time series forecasting models. This is partly true with the case studied here. Our small chain of independent bakeries operates three retail outlets, and each bakery produces its own sales data that over time provides a time series. Therefore, relevant forecasting models can be applied and offer potentially accurate forecasts of sales that will in turn offer the business the opportunity to make more targeted orders of foods which in turn will lead to less wastage at the end of the day.

The graphs below show the fluctuation of data in terms of sales of coffees and that of sandwiches per day across all branches of the bakery chain.

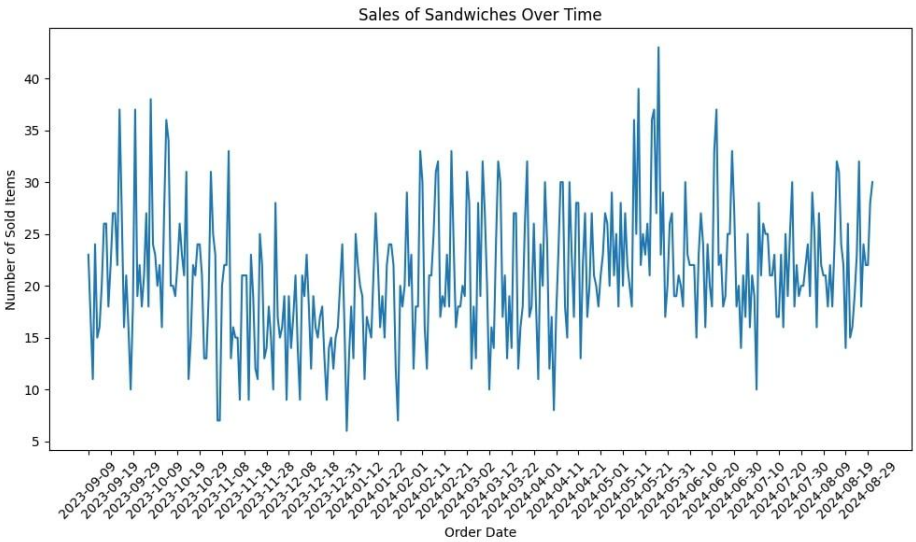

Fig. 1. Sales of Sandwiches over time – all bakeries

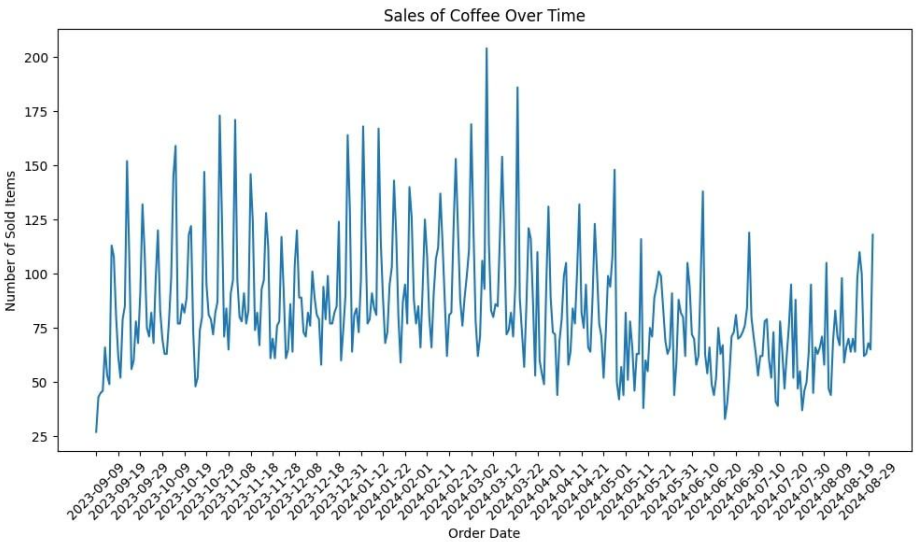

Fig. 2. Sales of Coffee over time – all bakeries

As can be easily seen the two sets of data yielding the above graphs vary considerably as to their behaviour, and the range of values. This suggests that different models will be required to predict future sales of these two groups of products. This is further emphasized as the nature of the two products is different. Coffees are made at the premises and on demand, while their raw materials have a much longer shelf life. On the

contrary sandwiches require preparation away from the point of sales and both the raw materials used, as well as the product have a much shorter shelf life of usually a day. A further challenge is that the volume of sandwiches sold is rather low and thus predictions as to expected sales can be more difficult.

2.1 Methodology

The sales data was obtained from a system used to process the sales in each of the outlets. This allowed data to be collected per different products for each of the three outlets. Given the challenges due to low volumes of sales of different types of foods the research focused on experimenting with the most popular product as sold at the busiest of the three outlets. This was further justified by the fact that each outlet has a different business model, dictated by each location and the type of clientele it attracts. Furthermore, some outlets might not have all the food items that are typically found in other outlets. In forecasting we are particularly keen to explore seasonality, i.e. repeating short-term fluctuation, and trend, long-term change in the time series model.

Prophet has been chosen as the forecasting model for this work. Its benefits include ability to add holiday effect by country/or even state easily, as special events/holidays have impact on sales, so by incorporating them into the model that can differentiate between normal days versus special events/holidays. Furthermore, Prophet can handle missing data as well as outliers automatically. It also can handle time series with complex seasonality, something that is relevant to our case here since each outlet has its own business characteristics and constraints. It is easy to use and is open source [6].

3 Related Work

The business world nowadays considers accurate prediction of sales as an essential task that can directly impact their ability to formulate effective plans, particularly in response to changing market trends, while also enabling them to optimize inventory management. An added advantage is normally enhanced customer satisfaction through timely product availability and possibly better prices through passing on part of the savings incurred through better utilization of business resources, to the customers. Most historic sales data is obtained in the form of time series and literature offers abundant choice of models for providing time series analysis. The authors in [7] highlight the importance of understanding the variations within time series data, particularly the influence of seasonal patterns, something that is directly relevant to bakeries as to the choice of products that could be offered, the expected volumes of sale, and timing of such sales related to weather, and other human controlled activities (such as festivals or sporting events) that occur regularly.

In [8] the author discusses constraints in relation to the data used in time series forecasting and while their work focuses on sales of alcoholic beverages in the USA, the are similarities between this work and ours presented here. Sales of alcohol present more significant fluctuations compared to other food categories and these induce more constraints. The authors performed machine learning with Prophet, an open-source

forecasting software package specializing in dealing time series released by Facebook and is much more effective in handling such data. This is particularly encouraging for our work as the data set used displays similarities in nature to our data. Our own data though has the further challenge of having less historic cycles as can be witnessed in the next section of this paper.

A further work whose data displays similar variations and constraints as ours is that discussed in [9]. The authors stress the importance of preprocessing data analysis to identify the dominant elements of the time series to ensure the most important contributing attributes of the data set are considered for processing. This corresponds with the approach adopted in our work and strengthens our confidence in the outcome.

Finally, the authors in [10] confirm the suitability of the use of Prophet as an appropriate method for our data analysis. Being mainly an extension of the generalized linear model using nonlinear smoothing functions. The advantage of the Prophet method over the diversity of statistical and time series forecasting methods such as autoregressive integrated moving average (ARIMA), seasonal autoregressive integrated moving average (SARIMA), fractional autoregressive integrated moving average (FARIMA), is the approach of the analyst-in-the-loop. This allows users to apply their domain knowledge about the dataset to the forecasting algorithm without having any knowledge of the statistical methods working from within. In our case the expertise of the business professional and future user is utilized to impact the development of the model with domain expert knowledge that would otherwise evade the span of expertise of the data analysts. This can have a very positive impact on the quality of the predictions obtained as it allows expert fine tuning that cannot be interpreted though the data.

4 Results and discussion

Following a preprocessing of that sales data over a period of twelve months the focus of the work turned to the highest selling products within the main two categories of the sales, mainly “Coffee Latte” and “Tuna Focaccia”. Despite the different nature of the products the sales data indicates similar behaviour with very few diversions from the mean value of sales instances daily, as shown in figures 3 and 4 below.

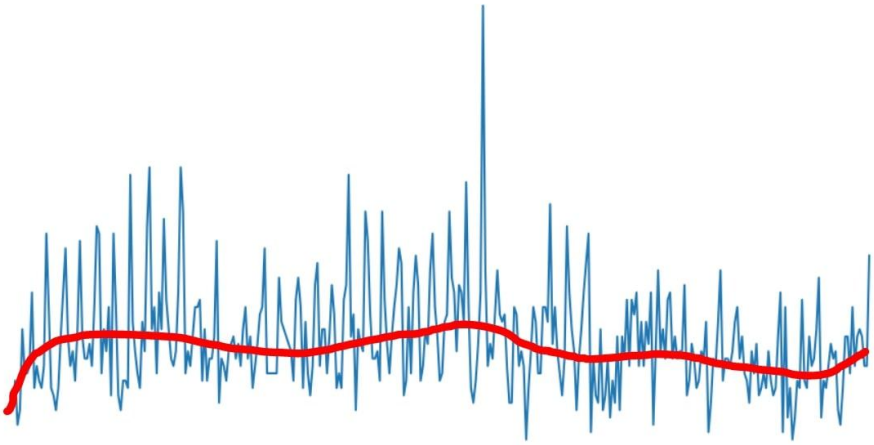

Fig. 3. Sales of Coffee Latte over time

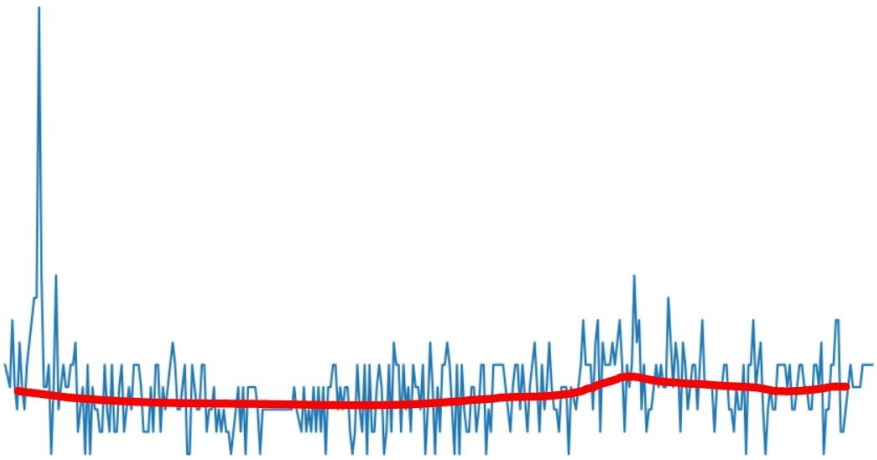

Fig. 4. Sales of Tuna Focaccia over time

To attempt to refine the series as shown in figures 3 and 4 above, we have used Prophet to add holiday effect, to explore the impact special events/holidays have can have on sales, so by incorporating them into the model can differentiate between normal days versus special events/holidays. The model was trained with no holiday effects first vs with holiday effect afterwards. The time of the data set was "Start Date is 2023-09-09 and End Date is 2024-08-31". The model was trained on 70% of the dataset and then it was utilized to predict the rest 30% of the values, which are then compared to the actual values for the remaining 30% of the original dataset.

4.1 The Prophet Forecasting Model

Prophet uses a decomposable time series model with three main model components: trend, seasonality, and holidays. They are combined in the following equation:

$$y(t)= g(t) + s(t) + h(t) + \varepsilon t \quad (1)$$

Where:

- g(t): piecewise linear or logistic growth curve for modeling non-periodic changes in time series
- s(t): periodic changes (e.g. weekly/yearly seasonality)
- h(t): effects of holidays (user provided) with irregular schedules
- εt: error term accounts for any unusual changes not accommodated by the model

MAPE - Mean Absolute Percentage Error was obtained

(the higher the value the worse the model is, the lower the value the better the model)

Mean Absolute Percentage Error (MAPE): 38.36% obtained is acceptable but not necessarily very good. This was achieved without considering the holiday effect on the time series for Coffee Latte

The next step involved incorporating the holiday effect with the MAPE improving by just under 2% (36.69%). Although this was an improvement, this was not a significant one. The next step though offered us the opportunity to explore the full value of Prophet by following Hyperparameters in Prophet and their effect:

```
from sklearn.model_selection import ParameterGrid
params_grid = {'seasonality_mode':('multiplicative','ad-
ditive'),
               'change-
point_prior_scale':[0.1,0.2,0.3,0.4,0.5,0.6,0.7],
              'holi-
days_prior_scale':[0.1,0.2,0.3,0.4,0.5,0.6,0.7],
              'n_changepoints' : [100,150,200,300,400],

'weekly_yes_no_':[True,False],

              }
grid = ParameterGrid(params_grid)
cnt = 0
for p in grid:
    cnt = cnt+1

print('Total Possible Models',cnt)
```

Total 980 Models

```
import random

model_parameters = pd.DataFrame(columns =
['MAPE','Parameters'])
for p in grid:
    test = pd.DataFrame()
    print(p)
    random.seed(0)
    train_model = Prophet(changepoint_prior_scale =
p['changepoint_prior_scale'],
                          holidays_prior_scale = p['holi-
days_prior_scale'],
                          n_changepoints = p['n_change-
points'],
                          seasonality_mode = p['seasonal-
ity_mode'],
                          weekly_seasonal-
ity=p['weekly_yes_no_'],
                          interval_width=0.95)
    train_model.add_country_holidays(country_name='UK')
    train_model.fit(Training_dataset)
    forecast = train_model.predict(forecast_date)
    df_eval = Testing_dataset.merge(forecast[['ds',
'yhat']], on='ds', how='inner')

    mape = mean_absolute_percentage_error(df_eval['y'],
df_eval['yhat']) * 100  # Convert to percentage

    print(f"Mean Absolute Percentage Error (MAPE):
{mape:.2f}%")
    print('Mean Absolute Percentage Error(MAPE)----------
--------------------------',mape)
    model_parameters = pd.concat([model_parameters,
pd.DataFrame([{'MAPE': mape, 'Parameters': p}])], ig-
nore_index=True)
```

We have managed to improve the model, from 36% to 30% via selecting the correct values for hyperparameters.
There are two key factors here.
Additive: states that season effect is increased by a constant rate no matter how the overall sales are going, e.g. if we have additive seasonality effect in our timeline, and

we sell ice cream, we expect to see that ice cream sales would increase EACH SUMMER by 50 units for example, even if the overall sales are terrible, but each summer our sales would increase by 50
Multiplicative: Relies more on the overall sales, every December, sales increase by a certain percentage (e.g., 20%). This means that if your sales are higher, the increase will be bigger. If your sales are lower, the increase will be smaller.

4.2 Looking at the accuracy graphically

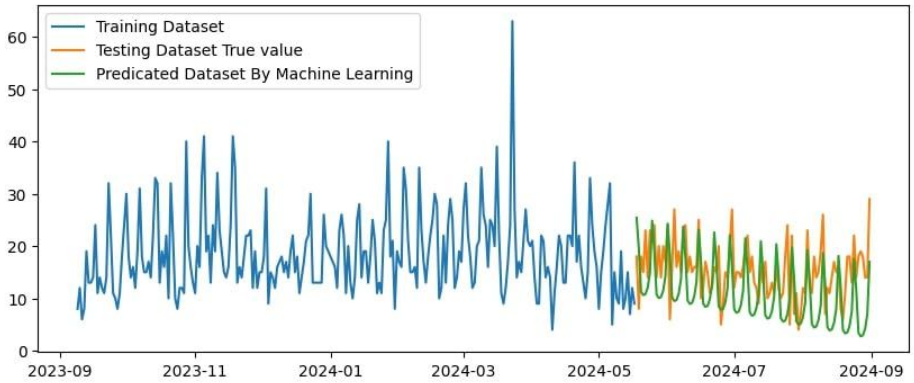

Fig. 5. Training Data Set Coffee Latte

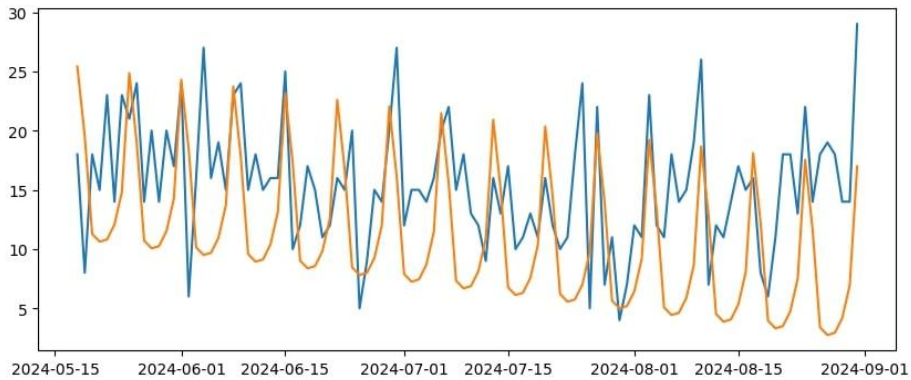

Fig. 6. Forecasting Coffee Latte Sales

Upon visual inspection of figures 5 and 6 it appears that the model has captured the general seasonality. The model's accuracy appears to decrease from 2024-08-15 onwards. This is possibly due to the nature of time series problems and indicates that the model may need to be updated regularly.

To explore room for further improvement the work considered the following options, (i)Sliding Window Approach and (ii)Rolling Forecast Origin.

For the sliding window approach the model is trained on a fixed-length window of past observations, when new data arrives, the oldest data point is removed, and the latest one is added (hence, the window "slides"). It is smaller compared to Rolling Forecast Origin, so training is quicker. One of the disadvantages is that it might lose some useful information from historical data, as it removes parts of the historic data over time.

Rolling Forecast Origin works similarly to Sliding Window, but we do not remove any data from our model, and when new data is provided, it is simply appended into the model. This could potentially enhance model accuracy, although it is computationally more expensive. For both approaches there is the assumption that there will be new data made available to update the model on a regular basis.

For the purposes of this work, it was decided to adopt Rolling Forecast Origin, starting with the model pretrained with 4 weeks of data, predicting the following week, getting next week's true values and append to the model, then trained the model on 5 weeks data, and so forth.

The Mean Absolute Percentage Error (MAPE) achieved was 34.96%

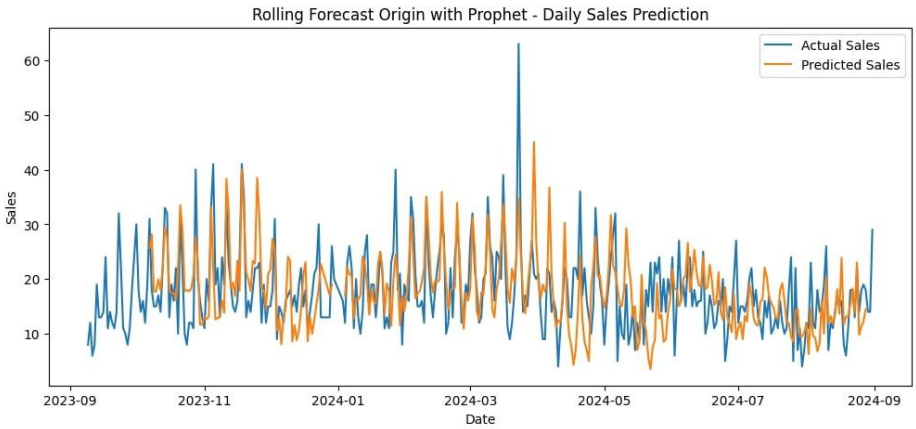

Fig. 7. Rolling Forecast Origin with Prophet – Daily Sales Prediction Coffee Latte

As seen in figure 7, the model appears to be more accurate than the previous one. The reason is that following a prediction for a week, once that week passes the actual data for that week, is used to update the model. Thus, the model keeps up with actual results as they evolve. Training is fine-tuned and accuracy is improved. Seeking possible ways to increase the accuracy of the rolling forecast origin, we increase the number of weeks the initial model was trained on, instead of 4 weeks, to 8 weeks and then apply rolling forecast.

The Mean Absolute Percentage Error (MAPE) improved to 31.66%. Figure 8 below shows the resultant graphical representation of the forecast. The graph demonstrates a good fit between actual and predicted levels of sales. There is certainly room for improvement, but the case presented and discussed here has its limitations and constraints that could limit the accuracy achieved. Considerable enhancements at the level of accuracy achieved have been demonstrated, despite a rather limited and possibly unstable

data set used. Further work is required to improve the desired level of accuracy in predicted sales. Such predictions can be enhanced by improving the consistency of the data set used and expanding the length of time for which data is available. Using Prophet provides a basis upon which user experience can interfere with the predictive model through a human expert and enhance the accuracy levels appropriately[11].

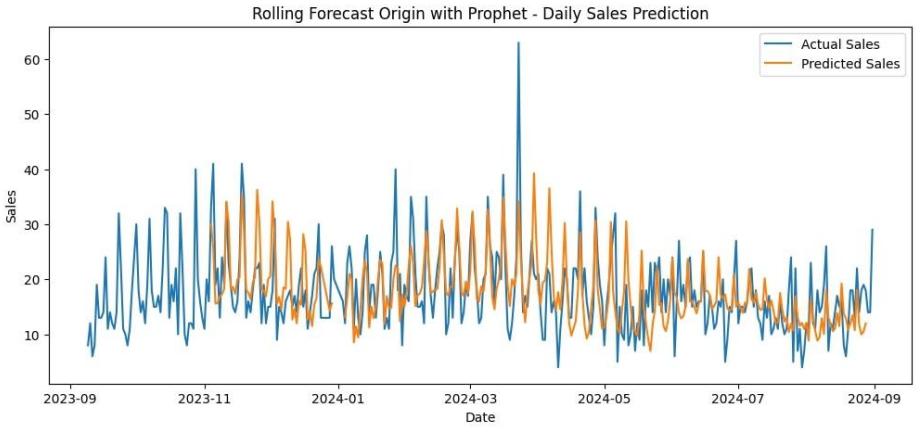

Fig. 8. Rolling Forecast Origin with Prophet – extended training period

5 Conclusions and further work

The work presented here has shown the potential of utilizing machine learning based tools to enhance traditional time series forecasting in successfully predicting future sales in bakeries. The quality of the data used here, and its consistency can affect the attainment of the desired outcome. Traditional time series forecasting has hidden biases and can lead to results that could yield large errors. Using an environment like Prophet can offer the benefit of enhancing a prediction by providing parameter selection in models that will improve the accuracy levels that are currently encountered.

The work is ongoing and further analysis with enhanced models, refined selection of datasets and other methods of enhancing results are being considered. Finally, the impact of capacity planning will be considered in attempting to enhance the accuracy of forecasting in a sensitive and very competitive business environment. This we expect to support effective profit margin improvements and the support of sustainability initiatives that counter food wastage.

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